Ecological Niche Modeling of *Vespa velutina*: Methodology, Theory, and Research Assets
Part I

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Introduction

Theoretical Framework: The Challenge of Modeling Invasive Niches

DEBE CONTENER:

1- INFO SOBRE LA VESPA VELUTINA Y EL POR QUÉ DE ESTE PROYECTO

2- INFO SOBRE EL PROYECTO EN SÍ, QUÉ HAREMOS Y CON QUÉ TRABAJAREMOS

3- LAS PRESENCIAS, ¿CÓMO LAS HEMOS REPRESENTADO?

Es posible que haya que añadir más variables, o hacer un PCA para ver cuales son más significativas o algo. Creo que el resto está relativamente bien.

4- LAS AUSENCIAS, ¿CÓMO LAS HEMOS REPRESENTADO?

Lo mismo que arriba

5- EL MODELO, ¿CÓMO LO HEMOS FINE TUNNEADO? ¿POR QUÉ ESTE MODELO?

6- MÉTRICAS

7- CONCLUSIONES Y TRABAJO FUTURO

The Species Distribution Models (SDMs) used in this study follow a correlative approach, inferring the environmental niche of *Vespa velutina* from its known presence locations. While mechanistic models rely on direct physiological measurements, correlative SDMs utilize the growing abundance of citizen science data to predict the species' potential spread. However, a significant source of uncertainty in these models is the inherent lack of "true absence" information. In an active invasion, a species may be absent from a locality not because the environment is unsuitable, but due to dispersal barriers, delayed detection, or biotic competition. Consequently, our methodology transitions from simple presence-only models—which often underestimate the "potential niche" by only reflecting the current occupancy—to a sophisticated presence-pseudoabsence framework. This allows us to move beyond a mere catalog of sightings and instead generate a robust risk assessment of the hornet's fundamental environmental threshold across the European landscape.

The distinction between the three modeling approaches lies in how they define the species' ecological boundaries and the data they require to do so. Simple presence-only models (e.g., BIOCLIM) are the most conservative, as they merely "wrap" a polygon around known sightings, making them prone to underestimating potential risk. Enhanced presence-only models (e.g., Maxent) improve upon this by comparing presence points against the environmental "background" of the study area; while more sophisticated, they are highly sensitive to sampling bias and do not explicitly model where the species is absent. Finally, presence-absence models (e.g., Random Forest) represent the most robust category for invasive species management. By utilizing pseudo-absences as a contrast, these models transform the ecological question into a binary classification problem (presence vs. absence), allowing the

algorithm to define sharp physiological thresholds and predict the species' potential niche rather than just its currently occupied territory.

¿CÓMO REPRESENTAN LAS PRESENCIAS EN EL PAPER?
(LEER UNO MÁS AL MENOS PARA COMPARAR Y MEJORAR)
Pues muy parecido a cómo lo he hecho yo, luego va a tocar explicarlo.

TEST Y CONTROL DE CALIDAD

To maximize the model's predictive reliability and ensure it captures the species' fundamental niche rather than just memorizing local spatial biases, we will implement a spatial cross-validation strategy inspired by the independent dataset approach. Instead of a simple random split of data, we will set aside a geographically and environmentally distinct region—such as the recent expansion front in Italy or a specific bioclimatic transition zone in the Iberian Peninsula—to serve as an independent 'blind' test set. The Random Forest model, refined by the SM4 protocol, will be trained on the remaining European occurrences. Its performance will then be evaluated by its ability to accurately project suitability onto this unseen territory. This approach, combined with a Principal Component Analysis (PCA) to ensure the training environmental space fully encompasses the test area's conditions, will allow us to rigorously quantify the model's transferability and its capacity to identify 'porous' corridors in regions it has not yet encountered. ESTO LO APLICAREMOS LUEGO EN EL TEST PARA VER COMO DE BUENO HA SIDO NUESTRO ENTRENAMIENTO

Hay parte en el pseudoabsence que debería de moverlo aquí:

HAY UN APARTADO DE REGULACIÓN Y ENCASILLAMIENTO QUE A NOSOTROS NO NOS INTERESA PORQUE GEE YA HACE EL TRABAJO POR NOSOTROS PARA RECALCULAR LOS POLOS!

1- e

Limpieza de datos

Modelo

Why not deep learning? (yet)

Feature engineering

Limpieza general

Generating pseudo-absences

Aquí hay que reordenar mucho content, está esparcido en la introducción y en el primer apartado del SM4.

To address the limitations of non-systematic data, three main strategies for pseudo-absence selection are traditionally employed. Simple random selection provides a broad environmental background but risks including suitable yet unoccupied sites, leading to model degradation. Geographically limited selection constrains points within a specific radius to focus on the species' dispersal range, though it often relies on arbitrary distance thresholds. Lastly, environmental profiling utilizes a two-step approach to select pseudo-absences only from areas identified as unsuitable through niche analysis, a method that has become a standard for increasing predictive accuracy. The SM4 protocol implemented in this study builds upon these techniques by integrating geographic constraints with environmental filtering, ensuring that pseudo-absences are both spatially relevant and ecologically distinct from the known presence niche. Therefore combining all three. The study also compares them (todo esto igual no es relevante o se puede acortar muchísimo)

Barbet-Massin basic model

To effectively model the potential niche of the *Vespa velutina*, the algorithm must distinguish between suitable and unsuitable habitats. This requires both presence and absence data. However, obtaining "true absences" (ground truth) is logistically challenging and expensive, because not sighting a hornet does not confirm its absence. Therefore, the standard practice is to generate pseudo-absences: synthetic points where the species is assumed to be absent.

While "presence-only" models like Maxent exist, literature consistently shows that presence-absence frameworks yield superior predictive performance. Nevertheless, generating these points is not a trivial task; simply scattering them randomly often fails to capture the species' complex invasive dynamics. To avoid arbitrary decisions, this project follows the guidelines established by the British Royal Society in the seminal paper: *'Selecting pseudo-absences for species distribution models: how, where and how many?'*.

A key takeaway from this research is that the sample size of pseudo-absences often outweighs the specific generation technique in terms of model impact. Furthermore, instead of using a single large pool of absences, the study suggests training in "batches" and averaging the results to minimize stochastic noise and improve accuracy. Since this project utilizes a Random Forest classifier, maintaining a 1:1 ratio (prevalence) is paramount. This balance, combined with a geographically and environmentally stratified selection, ensures the model learns the true biological limits of the species.

The following points outline the technical justification for each methodological choice based on the aforementioned study:

What is the optimal presence-to-absence ratio for Random Forest?

For Machine Learning algorithms like Random Forest, the optimal ratio is 1:1. According to Barbet-Massin et al., model accuracy (TSS) drops significantly when prevalence is low. Therefore, I have configured the dataset so the number of pseudo-absences matches the number of recorded nests (presences).

In fact, the study concludes that the quantity of pseudo-absences is, by far, the most influential factor in enhancing model precision for this algorithm.

How many times should pseudo-absence selection be replicated?

A single run is rarely optimal if the goal is to maximize model quality. To prevent random selection from "polluting" the results, I have implemented 5 replicates. The final map is an ensemble average of these runs, which enhances statistical robustness and filters out biases from random outliers.

The required number of replicates typically decreases as the number of pseudo-absences per batch increases. Given that our dataset is large (over 25,000 points) and we are using a 1:1 ratio, a small number of replicates is sufficient to reach a stable asymptote. This parameter is highly "fine-tunable," allowing for experimentation with 2, 3, or 5 replicates to determine which configuration yields the most consistent results.

What are the optimal number and weighting scheme for pseudo-absences per replicate?

For our Random Forest model, maximum accuracy is achieved when the number of pseudo-absences is approximately equal to the available presences. Given our robust dataset of ~25,000 observations, each replicate will contain exactly 25,000 pseudo-absences.

The study by Barbet-Massin et al. (2012) demonstrates that, unlike statistical models such as GLMs (which tolerate ratios of 1:100 or higher), Machine Learning algorithms like Random Forest suffer a drastic drop in sensitivity when an excessive number of absences is used. This occurs because the algorithm tends to overfit the majority class (absences), becoming unable to correctly identify high-risk areas.

Regarding weighting, the results indicate that the best performance is obtained by applying equal weights to both presences and pseudo-absences. Therefore, our ensemble strategy will average multiple replicates, each maintaining this critical 1:1 balance to ensure the Random Forest accurately learns the boundaries of the hornet's niche.

How do we choose where to place pseudo-absences?

For a Random Forest trained with a large number of presence points, the most effective method is Climate Exclusion (SRE - Surface Range Envelope). Selecting absences in areas that are environmentally distinct from presence locations significantly improves model accuracy compared to simple random sampling, which often risks generating "false absences" in potential expansion zones.

If we had fewer presence points, Geographic Exclusion (placing pseudo-absences at a specific spatial distance from known nests) would be more effective. This geographic approach could also be used to create a more "alarmist" model, prioritizing a higher rate of false positives over conservative estimates. However, given our high data volume, SRE ensures the model captures the true environmental niche.

What if my presence data is biased?

The Barbet-Massin study suggests that the number and weighting of pseudo-absences are largely unaffected by presence bias. However, if a significant climatic bias were present, geographic exclusion might be preferable over climatic exclusion; this prevents the model from assuming the species can only survive in the specific climates where it was sighted.

In our case, while sightings are heavily concentrated in temperate Atlantic climates, this is unlikely to be a sampling bias. *Vespa velutina* clearly favors high humidity, dense vegetation, and temperate conditions over the arid or extreme cold environments found in the Castilian Plateau, the Alps, or the Mediterranean coast.

With over 25,000 data points and a decade of invasion history, the absence of the hornet in these regions is not due to a lack of observers, it is due to a direct physiological barrier. While this factor has a smaller impact on overall precision than the total count of pseudo-absences, accounting for it ensures our model reflects biological reality rather than human observation patterns.

Conclusion

Ultimately, to maximize the predictive power of our Random Forest, we have implemented an ensemble strategy based on a strict 1:1 class balance and an equal weighting scheme. By combining the robustness of 25,000 records with a Climate Exclusion (SRE) method and averaging multiple replicates, we ensure a high-sensitivity model that respects the actual physiological limits of *Vespa velutina*. In the end, averaging several model runs is not just about statistical safety, it is what makes the final map realistic, ensuring it doesn't depend on the luck of where a few random points happened to fall.

Beyond Static Modeling: Implementing the SM4 Protocol

While our initial approach using the Barbet-Massin method provided a robust baseline, modeling the distribution of *Vespa velutina* requires a more surgical selection of pseudo-absences. Although the hornet has been present in Western Europe for over two decades, its expansion remains a dynamic process. The "sharp frontier" currently observed (particularly between the Atlantic regions and the

Castilian Plateau) might suggest a finished invasion; however, it is more likely a reflection of steep environmental gradients where the species faces immediate physiological barriers, such as extreme hydric stress and thermal limits.

The implementation of the SM4 (Four-Step Metadata-derived Multi-strategy) protocol is essential to address this nuance. Unlike broader methods, SM4 optimizes the spatial and environmental distance of pseudo-absences, ensuring they are not selected from "ecologically obvious" areas (which artificially inflates accuracy) nor from the species' immediate expansion front. This prevents the model from mistaking a temporary dispersal bottleneck for true habitat unsuitability.

By shifting to this multi-stage strategy, we expect to significantly reduce spatial autocorrelation and force the Random Forest to discriminate between truly habitable zones and those at the edge of the hornet's physiological threshold. This refinement allows us to determine if the current borders are absolute biological walls or merely "porous" corridors. **To ensure the highest methodological rigor, both techniques will be implemented and compared, evaluating whether the SM4 protocol provides a more realistic and high-resolution risk map compared to the established gold standard.**

SM1: Simple Random Selection

The Simple Random Pseudo-absence Selection (SM1) serves as the baseline approach in this study. In this method, pseudo-absence points are generated by randomly sampling locations across the entire study area, ensuring that they do not overlap with known presence records. While computationally straightforward, SM1 assumes that the species' absence is equally likely across all unsampled locations. However, as noted in the literature, this method is highly susceptible to sampling bias and may inadvertently include suitable but unoccupied habitats (false absences), potentially leading to a degradation in the model's ability to discriminate between presence and absence.

SM2: Spatially Constrained Selection

The Spatially Constrained Selection (SM2) introduces a geographic filter to the sampling process to account for the dispersal limitations of the species. Instead of sampling from the entire study extent, pseudo-absences are drawn only from a background area defined by a specific distance threshold—in this case, a 350 km radius around known presence points. By limiting the background to the species' 'reachable' geographic neighborhood, SM2 forces the model to learn the environmental characteristics that differentiate occupied sites from nearby unoccupied areas. This approach aims to reduce the noise introduced by geographically distant points that may be environmentally irrelevant to the current invasion front (of course there are not polar bears in the Sahara)

SM3: Environmental Profiling

The Environmental Pseudo-absence Point Selection (SM3) shifts the focus from geographic distance to ecological similarity. This method utilizes a two-step profiling procedure, specifically employing a One-Class Support Vector Machine (OCSVM) to define the species'

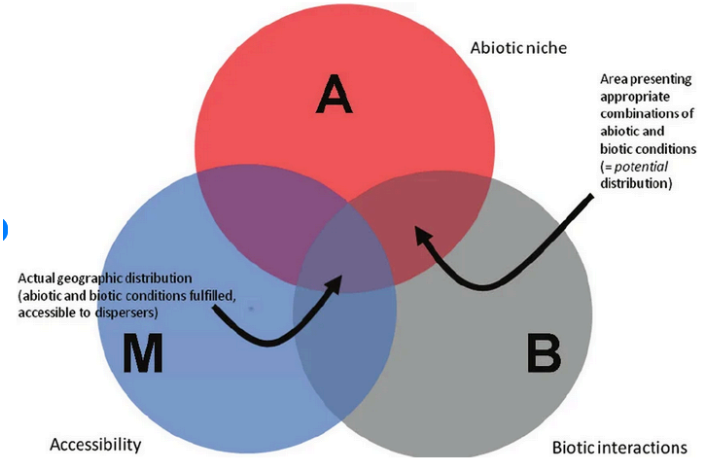
bioclimatic signature based solely on presence data. By training an ensemble of models, we identify areas of 'zero-similarity' where the environmental conditions diverge significantly from the known niche. Pseudo-absences are then selected exclusively from these unsuitable zones. This strategy ensures that the 'negative' training data are ecologically distinct from the 'positive' data, thereby minimizing the risk of including false absences in suitable but uncolonized regions.

Three Step Pseudo-absence Selection Method (SM4)

The selection of pseudo-absences is a critical stage in Species Distribution Modeling (SDM), as it defines the "background" against which the model learns to identify suitable habitats. To ensure high predictive accuracy and reduce spatial and environmental biases, this study also implements the SM4 protocol (Four-strategy Multi-stage). This method moves beyond simple random sampling by establishing a rigorous filtering process that balances geographic relevance with ecological contrast. By selecting pseudo-absences that are both spatially proximal to the invasion front and environmentally distinct from the species' known niche, the resulting Random Forest model is forced to identify the precise physiological thresholds that limit the expansion of *Vespa velutina*. The SM4 consists of 4 defined steps:

Step 1: Specifying geographical extent

The first stage of the SM4 protocol involves defining the geographic boundaries for the background data. Following the methodology optimized by recent research in invasive species modeling, a planar circular buffer of 350 km was established around each recorded presence point. This specific distance was selected based on variable importance analysis; empirical evidence shows that at a 350 km radius, the contribution of key bioclimatic variables to the model's first principal component is maximized.



Limiting the pseudo-absence pool to this extent serves two main purposes:

Dispersal Constraints: It focuses the model on areas that are geographically reachable by the species, representing the accessible area (the M in the BAM diagram).

Noise Reduction: It prevents the inclusion of redundant or irrelevant environmental data from distant regions (e.g., Northern Europe or arid Africa) that could obscure the critical drivers of the current invasion front in the Iberian Peninsula and France.

Within this 350 km constrained area, only the 1.4 km x 1.4 km cells not containing presence records were retained as candidates for the subsequent environmental filtering stages.

Step 2: Environmental profiling of background data

In the second stage, the background area (already constrained to 350 km in Step 1) undergoes an environmental profiling to filter out potentially suitable but unoccupied habitats. This study utilizes a

One-Class Support Vector Machine (OCSVM) classifier, an algorithm specifically designed to characterize the environmental niche using only presence data. The OCSVM identifies the complex, non-linear relationships between bioclimatic variables that define the survival threshold of *Vespa velutina*.

From the resulting probability surface, only locations with zero similarity (probability of 0) to the presence points are retained. This step is crucial for the model's integrity, as it ensures that pseudo-absences are selected from truly unsuitable environments. By excluding "false absences"—areas where the species could theoretically thrive but hasn't yet colonized—we prevent the Random Forest algorithm from receiving contradictory signals, thereby sharpening its ability to distinguish between habitable and inhospitable conditions.

Step 3: K-means clustering and representative sampling

The final stage addresses the potential for spatial and environmental redundancy within the pool of candidate pseudo-absences. To achieve a balanced and representative dataset, K-means clustering is applied to the zero-similarity locations identified in Step 2. The number of clusters (k) is set to match the total number of presence records available, ensuring a 1:1 presence-to-pseudo-absence ratio as recommended for balanced machine learning designs.

Rather than selecting points at random, t**he centroids of each cluster in the environmental feature space are selected as the final pseudo-absence locations.** This approach guarantees that the background data covers the entire range of unsuitable environmental conditions available within the study area. The projection of these centroids back into geographic space provides a robust, non-redundant set of points that maximize the model's ability to generalize and predict the invasion front with high confidence

MÉTRICAS DE COMPARACIÓN ENTRE TODOS LOS MODELOS (ESTO EN UN FUTURO SE TERMINARÁ)
(Este es el apartado final del paper)

EN PRINCIPIO NOSOTROS VAMOS A TENER 5 MODELOS: LAS MISMAS PRESENCIAS Y 5 DIFERENTES PSEUDOAUSENCIAS, TODAS ENTRENADAS CON UN RANDOM FOREST (JUSTIFICAR POR QUÉ SOLO USAMOS RANDOM FOREST):

El tipo de modelo (Random Forest, SVM, GLM...) no importa tanto como cómo eliges los datos.*The pseudo-absence selection method had a highly significant effect... but the interaction between model type and selection method was insignificant (PAPER SM4) (ESTO SIRVE PARA JUSTIFICAR QUE SOLO UTILICEMOS RANDOM FOREST)*

Data used for the SM4 paper -> con esto es suficiente?

PODRÍAMOS HACER ESTE MISMO ESTUDIO, DE CUÁNTAS VARIABLES UTILIZAMOS SEGÚN LA TÉCNICA DE PSEUDOAUSENCIAS.

CALCULAR AUC

CALCULAR DESVIACIÓN ESTANDARD

CALCULAR TTS

La conclusión del ANOVA y Tukey

…

Más cosas pero al final hacer un mapa todo guapo

No.	Variables	aaSM1	aaSM2	aaSM3	aaSM4	dvvSM1	dvvSM2	dvvSM3	dvvSM4
V1	Annual Mean Temperature	✓	✓	✓	✓		✓	✓	
V2	Mean Diurnal Range (Mean of monthly (max temp - min temp))							✓	
V3	Isothermality (P2/P7) (* 100)	✓	✓	✓		✓	✓	✓	
V4	Temperature Seasonality (standard deviation *100)	✓	✓	✓					
V5	Max Temperature of Warmest Month	✓		✓	✓			✓	
V6	Min Temperature of Coldest Month	✓		✓		✓		✓	
V7	Temperature Annual Range (P5–P6)	✓	✓						
V8	Mean Temperature of Wettest Quarter			✓					
V9	Mean Temperature of Driest Quarter	✓		✓					
V10	Mean Temperature of Warmest Quarter	✓	✓	✓	✓			✓	
V11	Mean Temperature of Coldest Quarter	✓	✓	✓		✓		✓	
V12	Annual Precipitation	✓	✓	✓	✓			✓	
V13	Precipitation of Wettest Month	✓	✓	✓	✓			✓	
V14	Precipitation of Driest Month	✓						✓	✓
V15	Precipitation Seasonality (Coefficient of Variation)	✓	✓		✓			✓	
V16	Precipitation of Wettest Quarter	✓	✓	✓				✓	✓
V17	Precipitation of Driest Quarter	✓	✓	✓	✓		✓	✓	
V18	Precipitation of Warmest Quarter						✓		
V19	Precipitation of Coldest Quarter	✓							
V20	Altitude	✓		✓	✓				
Total		17	11	14	8	3	4	13	2

*aa = *Aedes albopictus*, dvv = *Diabrotica v. virgifera*, SM1 = random pseudo-absence selection method, SM2 = spatially constrained random pseudo-absence selection method, SM3 = 2-step environmental profiling pseudo-absence selection method, SM4 = 3-step environmental profiling with spatial constraint pseudo-absence selection method.

Hay que integrar un gráfico en el que se vea la diferencia entre las casillas de presencia y de ausencia.

VAMOS A COMPARAR SM1, SM2, SM3, SM4 Y BARBET-BASSIN PARA VER SI TODO VA BIEN: TEST EN ITALIA, POLONIA O PORTUGAL.

HAY QUE SER MUY ORGANIZADO CON CÓMO GENERAMOS LAS PSEUDOAUSENCIAS Y COMO GUARDAMOS LOS DATOS

EXPLICACIÓN GENERAL DE CERCANÍA GEOGRÁFICA Y LEJANÍA AMBIENTAL (CHATGPT)

SM4 busca el "punto dulce" (sweet spot) entre dos mundos que a menudo chocan en la estadística:

Cercanía Geográfica: Si pones los puntos muy cerca, el modelo aprende a distinguir detalles sutiles del terreno, pero te arriesgas a que sean "falsas ausencias" (sitios donde la avispa podría estar mañana).

Lejanía Ambiental: Si pones los puntos muy lejos, el modelo tiene "certeza" de que ahí no vive la avispa, pero el Random Forest se vuelve perezoso porque la diferencia entre Galicia y Noruega es demasiado fácil de aprender.

Quehaceres:

- **Hay que nombrar los correlativos vs los mecanicistas, y hacer una introducción general a las pseudoausencias.**
- **Hacer por una parte un apartado de pseudoausencias simples, otro spatially constrained,**
- **Codear todo el planteamiento**

Librerías y recursos técnicos

Bioclim
Domain (vamos a checar qué es esto)

Papers útiles (Buffer con info aplicable a todos los demás apartados)

Niche modelling and landscape genetics of the yellow-legged hornet (*Vespa velutina*): An integrative approach for evaluating central–marginal population dynamics in Europe:

El paper nos dice que la variación genética es algo fundamental para predecir cómo de bien se va a expandir una especie, y que esta variación se ve muy muy disminuída en los bordes de la zona invadida por Vespa velutina. Los bordes de la expansión son genéticamente mucho menos diversas, porque menos individuos son capaces de adaptarse a ella. La diferencia ambiental también es fundamental para ver cómo evoluciona esta diversidad, como veremos a continuación.

¿Qué ocurre? Las especies invasoras son malas (expandir en esto),
Hay varios factores por los que una especie puede prosperar como invasora en un entorno extranjero, mientras que otros caen en desgracia.

Selecting pseudo-absences for species distribution models: how, where and how many?